

Project 3: Skin Lesion Classification for Melanoma Detection

Problem Statement

Develop a deep learning model to classify skin lesion images as either **benign** (non-cancerous) or **malignant** (cancerous). This is a critical task in computer-assisted dermatology.

Compare:

Transfer learning (ResNet50/MobileNetV3) fine-tuned

vs.

Feature extraction (pretrained CNN as frozen encoder) + classical ML (SVM/LogReg).

Core Concepts: Image Classification, Convolutional Neural Networks (CNNs), Transfer Learning, TensorFlow/PyTorch.

Dataset

You'll use the **HAM10000 ("Human Against Machine with 10000 training images")** dataset. It's a large collection of multi-source dermatoscopic images of common pigmented skin lesions.

Dataset download instructions:

1. Download the dataset from the [Harvard Dataverse](#).
2. You will need the image files and the HAM10000_metadata.csv file.
3. The 'dx' column in the metadata file contains the diagnosis. You will map these diagnoses into two classes: 'benign' and 'malignant'. (e.g., 'mel' and 'bcc' are malignant; others like 'nv' and 'bkl' are benign).

Starter Code

This snippet (provided separately) demonstrates how to set up a basic image classification model using **TensorFlow/Keras** with transfer learning from MobileNetV2.

Overall Guidance

1. **Data Preparation:** Load the metadata and create a balanced dataset. The original dataset is highly imbalanced. Use techniques like over-sampling the minority class (malignant) or under-sampling the majority class (benign).
2. **Image Augmentation:** Use ImageDataGenerator (Keras) or Transforms (PyTorch) to apply random rotations, flips, and zooms to the training images. This helps the model generalize better and prevents overfitting.
3. **Transfer Learning:** Use a pre-trained CNN like **MobileNetV2**, **ResNet50**, or **EfficientNet**. Freeze the initial layers and add your own classification head (a few Dense layers).

4. **Training & Evaluation:** Train the model and monitor its performance on a validation set. Evaluate the final model on a held-out test set using metrics like **Accuracy**, **Precision**, **Recall**, and the **AUC-ROC curve**. The ROC curve is particularly important for medical diagnostic tests.
5. **Classical ML comparison**
Extract global average pooled features from a frozen CNN and train an SVM; compare accuracy/F1 to fine-tuning.

Evaluation Rubric (100 points)

Criteria (Total 100)	Poor	Fair	Good
Translate business problem into technical problem, approach(10)			
Data Handling (10)	Fails to address class imbalance.	Attempts to balance data but method is flawed.	Effectively handles class imbalance; uses data augmentation properly.
Model Architecture (20)	Incorrect model implementation.	A basic CNN is used without transfer learning.	Correctly implements a transfer learning model with a proper architecture.
Performance & Evaluation (15)	Low accuracy and poor AUC.	Moderate performance; evaluation is limited to accuracy.	High accuracy and AUC; comprehensive evaluation with ROC curve analysis.
Compare classical ML Vs Fine Tuning (10)			
Code & Report (15)	Disorganized code, no report.	Functional code, minimal documentation.	Clean, well-documented code with a clear report explaining the methodology.
Team Work (10)	Each member working in silo, no coordination	Collaboration is happening but some gaps such as task division not proper	Well oiled team working effectively and optimally with each person contributing equally and effectively
Presentation (10)			